L9: The Fallacy of The Basic Reproductive Number

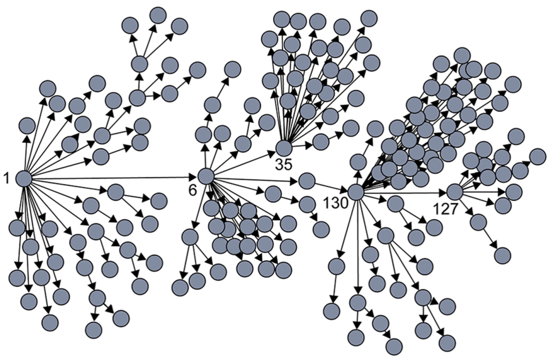

***Image Source:** Super-spreaders in infectious diseases [Richard A.Stein (Links to an external site.)](https://www.sciencedirect.com/science/article/pii/S1201971211000245) (https://www.sciencedirect.com/science/article/pii/S1201971211000245) , International Journal of Infectious Diseases, August 2011.*

It is important to realize however that R_0 is only an average – it does not capture the heterogeneity in the number of contacts of different individuals *(and it also does not capture the heterogeneity in the "attack rate" or "shedding potential" of the pathogen at different individuals)*. As we know by now, contact networks can be extremely heterogeneous in terms of the degree distribution, and they can be modeled with a power-law distribution of *(theoretically)* infinite variance. Such networks include hubs – and hubs can act as "superspreaders" during outbreaks.

SARS *(Severe Acute Respiratory Syndrome)* was an epidemic back in 2002-3. It infected 8000 people in 23 countries and it caused about 800 deaths. The plot shown here shows how the infections progressed from a single individual *(labeled as patient-1)* to many others. Such plots result from a process known as "contact tracing" – finding out the chain of successive infections in a population.

It is important to note the presence of a few hub nodes, referred to as ***superspreaders*** in the context of epidemics. The superspreaders are labeled with an integer identifier in this plot. The superspreader 130, for example, infected directly dozens of individuals.

The presence of superspreaders emphasizes the role of degree heterogeneity in network phenomena such as epidemics. If the infection network was more ***"Poisson-like",*** it would not have superspreaders and the total number of infected individuals would be considerably smaller.

Superspreaders in Various Epidemics

Disease	Location (year)	(R_0)[a]	SSE[b]	References
Ebola	Congo (1995)	1.83	21+,28-38	[85,86]
Measles	Greenland (1951)	16	250	[15,87]
	US (1985)	16	69,84	[15,88]
	Canada (1946)	16	678	[15,17]
PNEUMONIC PLAGUE	China(1946)	1.3	32	[89,90]
SARS	Hong Kong (2003)	3	187	[91,92]
	Vietnam (2003)	3	20	[91,93]
	Singapore (2003)	1.6	12,21,23,23,40+	[14,19]
	Canada (2003)	3	19,12-24	[91,94]
SMALLPOX	Yugoslavia (1975)	5.5	38	[95,96]

***Table Source:** [Cellular Superspreaders: An Epidemiological Perspective on HIV Infection inside the Body](https://doi.org/10.1371/journal.ppat.1004092%C2%A0) (https://doi.org/10.1371/journal.ppat.1004092%C2%A0) Kristina Talbert-Slagle et al., 2014,*

The table above confirms the previous point about superspreaders for several epidemics.

The third column shows R_0 while the fourth column shows ***"Superspreading events"*** *(SSE).* These are events during an outbreak in which a single infected individual causes a large number of direct or indirect infections. For example, in the case of the 2003 SARS epidemic in Hong Kong, even though R_0 was only 3, there was an SSE in which an infected individual caused a total of 187 infections *(patient-1 in the plot above).*

SSEs have been observed in practically every epidemic – and they have major consequences both in terms of the speed through which an epidemic spreads and in terms of appropriate interventions.

For example, in the case of respiratory infections *(such as COVID-19)* ***"social distancing"*** is an effective intervention only as long as it is adopted widely enough to also include superspreaders.